help create scripts that optimise load balancing, manage virtual machines, and ensure compliance with policies.

Troubleshooting: Generative AI can assist in diagnosing and resolving issues in cloud environments. By using prompts to guide AI through log analysis and error reports, it can identify the root cause of problems and suggest remedial actions, streamlining the troubleshooting process.

Popular GenAI tools for cloud platforms

Generative AI tools are available for major cloud platforms, each offering unique features to enhance cloud operations

Microsoft Azure

- ***Azure Cognitive Services:*** These provide AI capabilities including anomaly detection, which can be used for monitoring and troubleshooting.
- ***Azure Machine Learning:*** Enables the development and deployment of AI models that can automate cloud management tasks.
- ***Azure Functions and Logic Apps:*** We can write custom Azure functions or logic apps using Python, which can invoke OpenAI for specific resource handling purposes.

AWS (Amazon Web Services)

- ***Amazon SageMaker:*** Facilitates the creation and deployment of AI models for various operational tasks.
- ***Amazon CloudWatch:*** Integrates AI for monitoring and alerting, enhancing the efficiency of cloud operations.

Google Cloud

- ***Google AI Platform:*** Offers tools for building, deploying, and managing AI models that can be used for cloud operations.
- ***Google Cloud Operations Suite:*** Incorporates AI for monitoring, logging, and error reporting.

Cloud-agnostic tools

Several tools are designed to work across various cloud platforms, providing flexibility and consistency in cloud operations.

Terraform: An infrastructure-as-code (IaC) tool that allows for automated provisioning and management of cloud resources.

Kubeflow: Facilitates the deployment of machine learning models on Kubernetes, applicable across different cloud environments.

Ansible: An automation tool for configuration management and orchestration, supporting multiple cloud providers.

AIOps versus generative AI for cloud operations

AIOps (artificial intelligence for IT operations) and generative AI for cloud operations serve distinct purposes. AIOps focuses on using AI to analyse large volumes of data generated by IT environments to predict issues, automate processes, and improve overall performance. Generative AI, on the other hand, specialises in generating content and responses based on input prompts, making it ideal for scripting and automating specific tasks. Table 1 compares both.

Art of prompting in cloud operations

The 'art of prompting' refers to the skill of crafting precise, effective, and context-aware instructions or queries to interact with AI systems, automation tools, or cloud platforms to achieve desired outcomes efficiently. In the context of cloud operations, prompting is critical for managing resources, automating workflows, troubleshooting issues, and optimising performance through natural language interfaces or command-line tools integrated with AI or cloud APIs. A well-crafted prompt minimises ambiguity, reduces errors, and accelerates problem resolution or task completion.

In cloud operations, the art of prompting often involves interacting with AI-driven tools (like chatbots for cloud management), scripting automation, or querying monitoring systems. It requires an understanding of the system's language, the context of the operation, and the expected output.

The key principles of the art of prompting in cloud operations are:

Clarity and specificity: Ensure the prompt is clear and specific to avoid misinterpretation by the system or AI.

Contextual awareness: Include relevant details about the environment, service, or issue to narrow down the response or action.

Iterative refinement: Refine prompts based on initial outputs to achieve better results.

Action-oriented language: Use commands or queries that directly map to desired actions or insights.

Here are some real-world examples of prompting in cloud operations.

Table 1: AIOps vs generative AI for cloud operations

Aspect	AIOps	Generative AI
Primary function	Data analysis and process automation	Content generation and task automation
Usage	Predictive maintenance, anomaly detection, performance optimisation	Scripting for monitoring, management, troubleshooting
Tools	Splunk, Moogsoft, BigPanda	OpenAI GPT, Google AI Platform, Amazon SageMaker
Integration	IT service management platforms	Cloud management and automation scripts